

Design Verification Plan and Report (virtual-test version) — VBF-T5R1NOCEILING-DVPR-01

Cell: R4A MarginAnode1p0. One row per verification item; determinations are mechanical vs the log.jsonl entry-0 contract criteria; uncovered conditions honestly marked N/A.

Item	Condition	Result value	Determination	Source
1C discharge capacity	run-pyamm 1C_discharge, DFN, 298.15 K ambient	9.6251 Ah; terminates cleanly at 2.5000 V in 6930 s (no 7200-s truncation)	reported (no capacity threshold in contract)	cell/r4_A_1c_dfn.json; cell/r4_A_calc_dfn.json:capacity_ah
Energy density	calc-energy contract formula (energy Wh / mass kg; mass = sum layer thickness x (1-porosity) x density x area, electrolyte excluded)	583.93 Wh/kg (SPMe cross-check 583.46)	PASS (≥ 500.94)	cell/r4_A_calc_dfn.json:energy_density_wh_kg
4C fast-charge temperature rise	run-pyamm 4C_charge_45C, DFN, lumped thermal, 318.15 K ambient	T_max 320.36 K (47.2 degC)	PASS (≤ 333.15 K / 60 degC)	cell/r4_A_4c_dfn.json:T_max_K
4C fast-charge plating	run-pyamm 4C_charge_45C + --plating, DFN	anode surface potential min +0.0212 V (charge ends at 4.2 V ceiling, 84.3% SOC)	PASS (no plating; min > 0 V)	cell/r4_A_4c_dfn.json:anode_potential_v
Voltage window	parameter set lower/upper cut-offs	2.5 - 4.2 V	consistent with protocol termination conditions	parameter set
Nail penetration	-	N/A (beyond pure simulation boundary, requires physical experiment)	N/A	-
Overcharge to thermal runaway	-	N/A (beyond pure simulation boundary, requires physical experiment)	N/A	-
Crush / drop	-	N/A (beyond pure simulation boundary, requires physical experiment)	N/A	-
Cycle life	-	N/A (no aging model; aging protocol not part of this contract)	N/A	-
Rate-pulse (HPPC) internal resistance	-	N/A (beyond pure simulation boundary); note calc-energy reports quasi-static DCR 0.00109 Ohm	N/A	cell/r4_A_calc_dfn.json:dcr_ohm

Conclusion

3/3 contract criteria PASS at DFN, with SPMe cross-check agreeing on all three (ED 583.46; anode min +0.0270 V; T_max 320.44 K). Safety margins: anode surface +21 mV above plating onset; temperature 12.8 K below the 60 degC limit; ED 16.6% above the 500.94 Wh/kg floor. Uncovered items (nail, overcharge-to-thermal-runaway, crush, drop, cycle life, HPPC) are outside the pure-simulation boundary and are cited as limitations.